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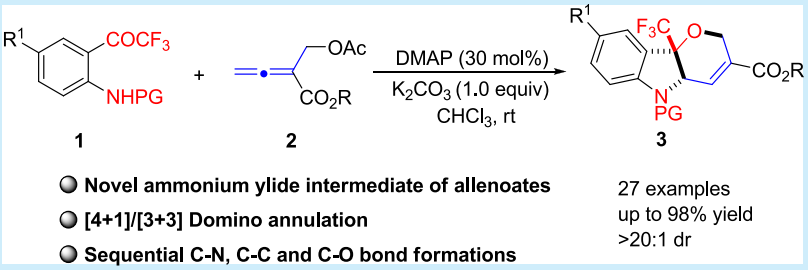
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Construction of CF_3-Containing Tetrahydropyrano[3,2-*b*]indoles through DMAP-Catalyzed [4+1]/[3+3] Domino Sequential Annulation

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ABSTRACT: A [4+1]/[3+3] domino sequential annulation reaction of *o*-aminotrifluoroacetophenone derivatives and *β*′-acetoxy allenoates enabled by DMAP has been reported. A variety of CF_3-containing tetrahydropyrano[3,2-*b*]indoles were obtained as a single diastereomer in high yields (≤98%) under mild conditions. The reaction can build one C−N bond, one C−C bond, and one C−O bond sequentially in a single step. The synthetic utility was demonstrated with gram-scale reactions and various transformations of the products.

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The pyranoindole skeletons are of great importance because of their wide distribution in classic human drugs that have diverse pharmacological properties (Figure 1).

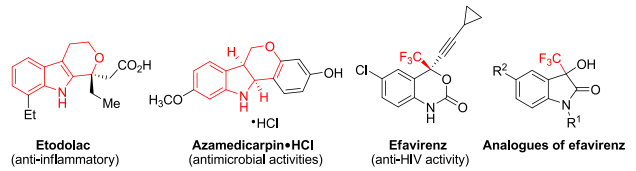


Figure 1. Examples of pharmaceuticals featuring pyranoindoles and CF_3-containing heterocyclic scaffolds.

For example, as a best-selling anti-inflammatory drug, Etodolac containing a pyranoindole core is widely used in the clinical treatment of arthritis and acute pain.[1] On the contrary, the introduction of fluorine-containing groups into drug molecules can improve molecular properties,[2] such as metabolic stability, lipophilicity, and permeability, and their interaction with biological targets, which has gradually become a common method for drug screening. Among them, CF_3-containing heterocycles have become some of the most promising fragments for drug molecules.[3] For example, Efavirenz, a molecule with a trifluoromethyl group in a heteroaliphatic ring, is a non-nucleoside reverse transcriptase inhibitor used in the treatment of patients with HIV.[4] Therefore, the development of simple and efficient synthetic methods for the synthesis of heterocycles containing CF_3 is important.

Since the pioneering work of Lu,[5] Lewis base-catalyzed annulation reactions of allenes have shown their capability in the construction of different ring system carbo- and heterocycles.[6] Among these cyclization reactions, nucleophilic phosphine catalysis has proven to be one of the most practical and powerful in domino annulation reactions,[7] and multiple rings can be constructed in a one-step reaction to generate molecules that are structurally diverse and complex.[8] In contrast, amines are generally less nucleophilic than similarly substituted phosphines;[9] therefore, tertiary amine-catalyzed domino annulations are relatively rare,[10] and most of them can construct only a single ring in one step. In 2010, *β*′-acetoxy allenoate **2a** was disclosed by Tong;[11] after the phosphine attack, the acetate group at the *β*′ position undergoes S_N2' displacement, creating electrophilic phosphonium diene **A**, whose *γ* and *β*′ positions can be attacked by nucleophiles and complete the subsequent cyclization reactions (Scheme 1a).[12] Similarly, under the catalysis of tertiary amines, the

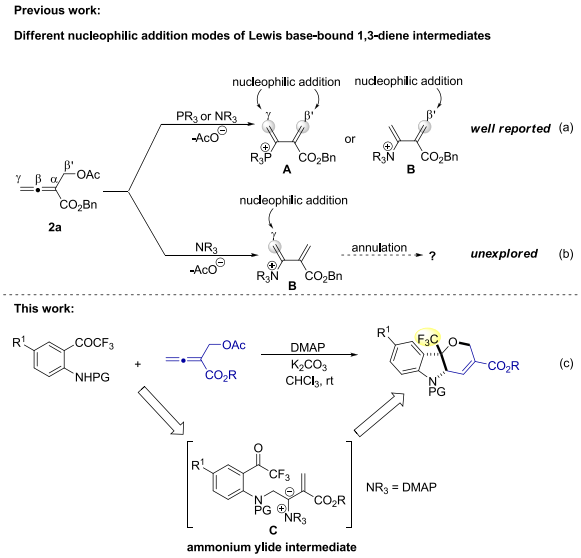
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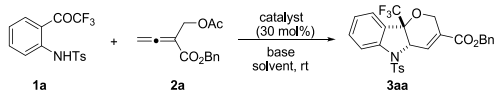
Scheme 1. Design Plan for the Lewis Base-Catalyzed Domino Annulation



corresponding electrophilic diene intermediate **B** can be generated, but to date, all nucleophiles are additions to its β' position (Scheme 1a).[13] There has been no report on the addition of nucleophiles to the γ position of intermediate **B**, for there is no empty orbital in the ammonium ion of intermediate **B** to stabilize the carbon anion in the subsequently generated ammonium ylide (Scheme 1b).[14] To overcome the limitations, and on the basis of our group's previous work in the exploration of novel reaction modes of β'-acetoxy allenoate[15] and sequential annulations in organocatalysis to construct the polycycles,[16] we envisioned that the DMAP (4-dimethylaminopyridine) would provide a π-system and make it a more stable corresponding ylide intermediate through a p-π conjugation between the carbon anion and the aromatic ring of DMAP. Herein, we sought to use *o*-aminotrifluoroacetophenone derivatives as electron donor−acceptor compounds to realize a DMAP-catalyzed [4+1]/[3+3] domino annulation for the efficient formation of a CF_3-containing tetrahydropyrano[3,2-*b*]indole system (Scheme 1c). One C−N bond, one C−C bond, and one C−O bond were sequentially built in a single step. To the best of our knowledge, the introduction of CF_3 unit into pyranoindoles remains a blank field.

To examine the feasibility of the envisaged domino reaction, we chose *o*-aminotrifluoroacetophenone derivative **1a** and allenoate **2a** as the model substrates to optimize this reaction in chloroform at room temperature in the presence of Cs_2CO_3. We initiated our research by examining various Lewis bases as catalysts (30 mol %). In the presence of PPh_3 or tertiary amine DABCO, no desired product **3aa** was observed (Table 1, entries 1 and 2). To our delight, DMAP was found to be an effective catalyst, furnishing product **3aa** in 44% yield with >20:1 dr (Table 1, entry 3). The blank control experiment showed that without DMAP as a catalyst, the desired cyclization process could not be achieved (Table 1, entry 4). We further investigated a range of inorganic and organic bases (entries 5−8). It was established that the utility of K_2CO_3 enhanced the yield of **3aa** to 61% (entry 6), and the yield was also lower in the absence of a base (entry 9). The examination of the solvent effect revealed that chloroform was the suitable

Table 1. Optimization of the Reaction Conditions[a]



entry	catalyst	base (equiv)	solvent	yield (%)[b]	dr[c]
1	PPh_3	Cs_2CO_3 (1.2)	$CHCl_3$	0	–
2	DABCO	Cs_2CO_3 (1.2)	$CHCl_3$	0	–
3	DMAP	Cs_2CO_3 (1.2)	$CHCl_3$	44	>20:1
4	–	Cs_2CO_3 (1.2)	$CHCl_3$	0	–
5	DMAP	Na_2CO_3 (1.2)	$CHCl_3$	57	>20:1
6	DMAP	K_2CO_3 (1.2)	$CHCl_3$	61	>20:1
7	DMAP	Et_3N (1.2)	$CHCl_3$	48	>20:1
8	DMAP	CH_3ONa (1.2)	$CHCl_3$	trace	>20:1
9	DMAP	–	$CHCl_3$	45	>20:1
10	DMAP	K_2CO_3 (1.2)	acetone	28	>20:1
11	DMAP	K_2CO_3 (1.2)	THF	35	>20:1
12	DMAP	K_2CO_3 (1.2)	toluene	11	>20:1
13	DMAP	K_2CO_3 (1.0)	$CHCl_3$	64	>20:1
14[d]	DMAP	K_2CO_3 (1.0)	$CHCl_3$	73	>20:1
15[e]	DMAP	K_2CO_3 (1.0)	$CHCl_3$	68	>20:1
16[d,f]	DMAP	K_2CO_3 (1.0)	$CHCl_3$	58	>20:1
17[d,g]	DMAP	K_2CO_3 (1.0)	$CHCl_3$	48	>20:1
18[d,h]	DMAP	K_2CO_3 (1.0)	$CHCl_3$	71	>20:1

[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.2 mmol, 2.0 equiv), catalyst (30 mol %), and base in 1.0 mL of the indicated solvent at room temperature for 18 h. [b]Isolated yield. [c]Diastereoselectivity was determined by ^{1}H NMR spectroscopy. [d]Solvent volume of 2.0 mL. [e]Solvent volume of 4.0 mL. [f]With 1.5 equiv of **2a**. [g]Reaction carried out at 0 °C. [h]Reaction carried out at 40 °C.

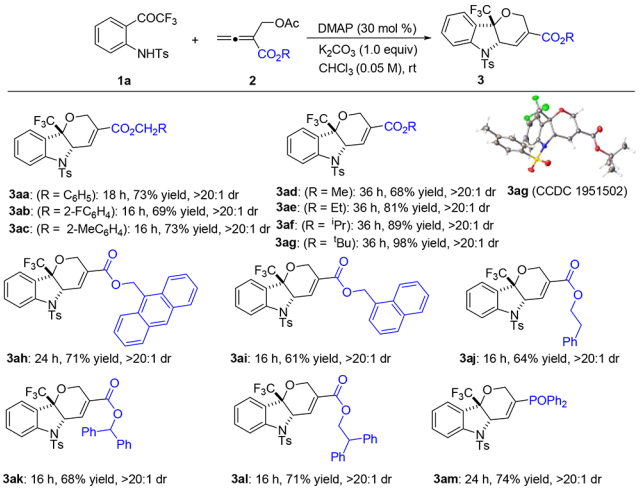
medium for this transformation (Table 1, entries 10−12). Then, the use of a lower base loading can slightly increase the yield to 64% (Table 1, entry 13). Surprisingly, decreasing the concentration of the system to 0.05 M provided a much cleaner reaction, and the yield of **3aa** was further improved to 73% (Table 1, entry 14). Finally, further decreasing the reaction concentration, decreasing the amount of **2a**, and changing the reaction temperature did not provide better results (Table 1, entries 15−18).

With the optimized reaction conditions in hand, the substrate scope of allenoates was then tested (Scheme 2). In addition to **3aa**, electron-withdrawing (**3ab**) or electron-donating (**3ac**) substituents at the *ortho* position of the phenyl ring and heteroaryls such as 9-anthryl (**3ah**) and 1-naphthyl (**3ai**) were all suitable for the reaction, and the corresponding products were obtained as a single diastereomer (>20:1 dr) in good to high yields (61−73%). The R substituent can also be a diphenylmethyl group, delivering the desired product **3ak** in good yield. Apart from benzyl allenoates, allenoates containing alkyl substituents were also suitable. The alkyl groups can be linear (**3ad** and **3ae**), branched (**3af** and **3ag**), or aryl-bearing (**3aj** and **3al**), and the corresponding products were obtained in good to excellent yields (64−98%) with excellent diastereoselectivity (>20:1 dr). In addition, this methodology is also amenable to the use of diphenylphosphine oxide **2m**, providing the corresponding product **3am** in high yield. The structure and stereochemistry of **3ag** were unambiguously characterized by single-crystal X-ray analysis.[17]

Subsequently, the scope of *o*-aminotrifluoroacetophenone derivatives **1** was investigated (Scheme 3). Both electron-donating and -withdrawing substituents could be installed at the *para* position of the amino group of substrates **1**, with the

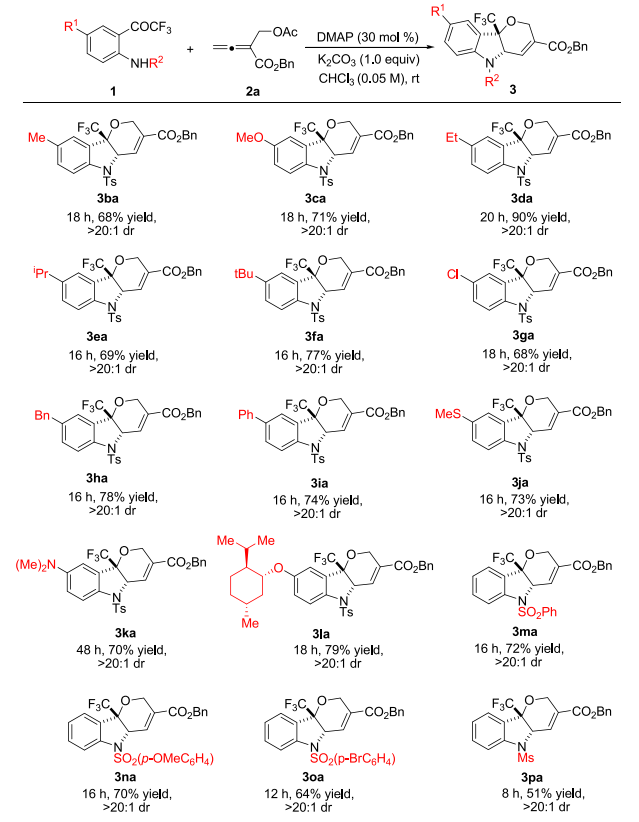
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Scheme 2. Substrate Scope for Reactions of 2 with 1a[a]



[a]The reactions were carried out with **1a** (0.1 mmol), **2** (0.2 mmol), DMAP (30 mol %), and K_2CO_3 (0.1 mmol) in 2.0 mL of $CHCl_3$ at room temperature. Diastereoselectivity was determined by ^1H NMR spectroscopy.

Scheme 3. Substrate Scope for Reactions of 1 with 2a[a]

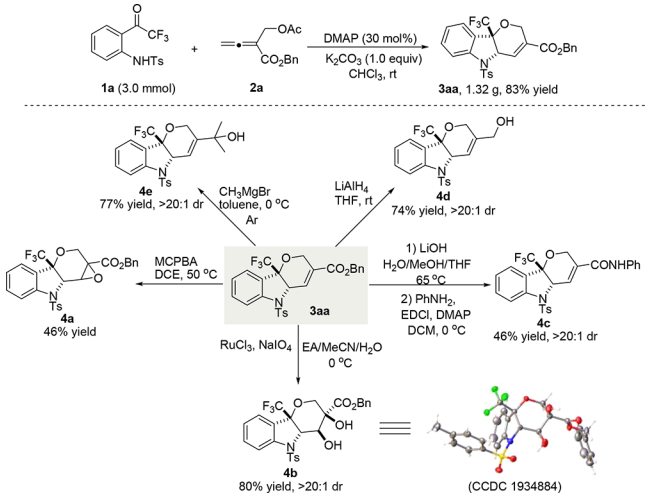


[a]The reactions were carried out with **1** (0.1 mmol), **2a** (0.2 mmol), DMAP (30 mol %), and K_2CO_3 (0.1 mmol) in 2.0 mL of $CHCl_3$ at room temperature. Diastereoselectivity was determined by ^1H NMR spectroscopy.

corresponding products afforded in good to high yields (68–90%) (**3ba**–**3ha**). Further investigation showed that substrates **1** bearing an aryl group (**3ia**) were also efficient for the transformation. Moreover, the aromatic rings of substrates **1** bearing other heteroatom substituents at the *para* position such as SMe, NMe_2, and L-menthyl did not interfere with the reaction efficiency (**3ja**–**3la**). Finally, variation of the phenyl ring of the arylsulfone moiety and methanesulfonyl on the amino group can also be tolerated, and products (**3ma**–**3pa**) were obtained in 51–72% yields. Notably, only a single diastereomer was observed in all cases.

To demonstrate the practical applicability of this domino annulation, a gram-scale reaction of **1a** was performed under standard reaction conditions (Scheme 4). The reaction still

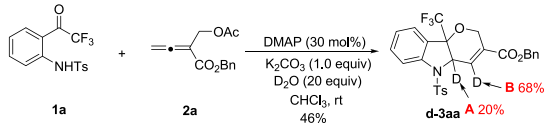
Scheme 4. Gram-Scale Synthesis and Further Transformation of the Product



smoothly proceeded to afford **3aa** with an improved yield (83%) and excellent diastereoselectivity. Subsequently, synthetic transformations of product **3aa** were conducted. As shown in Scheme 4, epoxidation and dihydroxylation readily converted **3aa** into **4a** and **4b** in 46% and 80% yields, respectively, with excellent diastereoselectivity. The structure and stereochemistry of **4b** were determined by single-crystal X-ray analysis.[18] Moreover, consequent hydrolysis/amidation of **3aa** enabled the construction of amide **4c** in 46% total yield. Facile reduction of the ester moiety with $LiAlH_4$ at room temperature furnished **4d** in high yield. In addition, treating **3aa** with PhMgBr afforded **4e** with a tertiary alcohol group.

To study the reaction mechanism, we conducted the domino reaction with **1a** and **2a** in the presence of 20 equiv of D_2O under the optimized reaction conditions (Scheme 5). The

Scheme 5. Deuterium Labeling Experiment

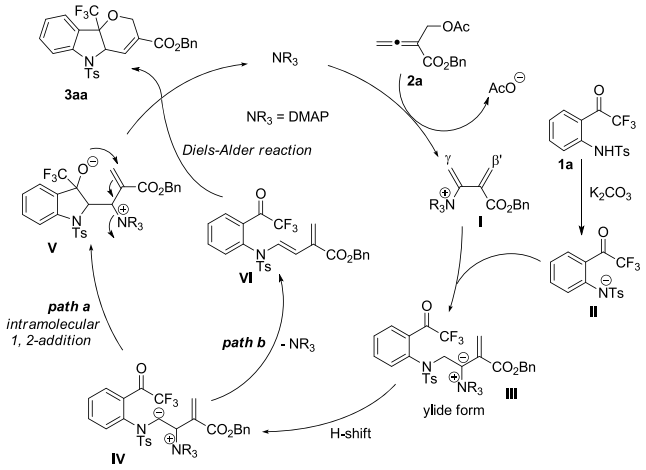


deuterated product d-**3aa** was obtained in 46% yield, and H/D exchange was observed at the β and γ positions of allenoate **2a**, indicating the possibility of the involved carbanion at position 2 of the allene intermediate.

On the basis of the control experimental (Table 1, entries 4 and 9) and the experimental result with deuterium, a plausible reaction pathway for this domino annulation is proposed in Scheme 6. The reaction is triggered by the addition of DMAP to allenoate **2a** and subsequent 1,2-elimination of the acetate

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Scheme 6. Plausible Reaction Mechanism



group to generate electrophilic diene intermediate **I**. Then, ammonium ylide **III** was formed by a γ-addition between intermediate **I** and intermediate **II** (deprotonated **1a**), induced by the electron-withdrawing ability of the ammonium in intermediate **I** and the p-π conjugation between the carbon anion and the aromatic ring moiety of the ammonium in ammonium ylide **III**. The subsequent 1,2-proton shift of **III** followed by intramolecular 1,2-addition in intermediate **IV** gives intermediate **V** (path a). Finally, intramolecular S_N2' substitution in intermediate **V** yields product **3aa** with DMAP catalyst regeneration. On the contrary, we thought that the 1,2-proton shift from intermediate **III** to **IV** lacks a certain driving force, so another plausible pathway was proposed. On the basis of intermediate **IV**, DMAP is removed by 1,2-elimination to form intermediate **VI** (path b), and then an intramolecular Diels–Alder reaction in intermediate **VI** occurs, affording product **3aa** directly. However, we failed to isolate intermediate **VI** during the whole reaction. The observed excellent diastereoselectivity of the reaction indicated that the concerted Diels–Alder reaction pathway would be more possible.

In summary, we have developed a novel [4+1]/[3+3] domino sequential annulation reaction between *o*-amino-trifluoroacetophenone derivatives and β'-acetoxy allenoates by employing the DMAP catalyst, and a series of CF_3-containing tetrahydropyrano[3,2-*b*]indole products were synthesized in high to excellent yields with a single diastereoselectivity under mild reaction conditions. Moreover, the mechanistic studies revealed that the reaction proceeds through an ammonium ylide intermediate. The gram-scale reactions and the various transformations of the products may have broad applications in the synthesis of valuable pharmaceuticals with trifluoromethyl-substituted heterocycles. Further investigations of the reaction mechanism and the enantioselective version of this reaction are currently underway in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c02164.

Experimental details, characterization data for new compounds, copies of NMR spectra, and X-ray crystal structures (PDF)

Accession Codes

CCDC 1934884 and 1951502 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(17) CCDC 1951502 (**3ag**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

(18) CCDC 1934884 (**4b**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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